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ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.*(An ISO 9001:2015 Certified College)***DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING****B. TECH DEGREE FIRST INTERNAL TEST- JANUARY 2022****FOURTH SEMESTER – Batch A&B****CST 205 : OBJECT ORIENTED PROGRAMMING USING JAVA**

Time: Hours

Max. Marks: 50

PART A

Answer all questions (3x5=15 marks)

Qn. No.	Questions	Marks	K level	CO & KL
1	Discuss the concept of classes and objects in Java language using an example of a 'student' object in a Student management application	3	K2	CO1&K2
2	Outline the reasons as of why Java programs are known as "Write Once and Run Anywhere" programs?	3	K2	CO1&K2
3	List the advantages of using packages in Java programming. How do you create and import a package in Java?	3	K2	CO2&K2
4	Write a java program to check if a Given Number is Perfect Square using user defined Logic.	3	K3	CO2&K2
5	With a flowchart illustrate and explain the flow of control in the execution of try-catch-finally statement	3	K2	CO3&K2

4/2
9/2

PART B

Answer any five questions (7x5=35 marks)

Qn. No.	Questions	Marks	K level	CO & KL
6	Create a class 'Bank' with a method 'getBalance' which returns 0. Make its three subclasses named 'BankA', 'BankB' and 'BankC' with a method with the same name 'getBalance' which returns the amount deposited in that particular bank. Call the method 'getBalance' by the object of each of the three banks.	7	K3	CO1&K2
7	Describe buzzwords of Java that defines the Java programming language.	7	K2	CO1&K2
8	Illustrate the Java Programming and Runtime Environment. Explain the roles of each component of it while compiling and executing a java program.	7	K1	CO1&K2

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9	Create an abstract class 'Parent' with a method 'message'. It has two subclasses each having a method with the same name 'message' that prints "This is first subclass" and "This is second subclass" respectively. Call the methods 'message' by creating an object for each subclass.	7	K3	CO2&K2
10	List out any three literal types in Java. Give examples for each. Discuss about relational and conditional operators in Java with examples and compare its precedence.	7	K2	CO2&K2
11	Write a java program to take 20 integer inputs (+ve ,-ve and 0) from user and print the following: number of positive numbers number of negative numbers number of odd numbers number of even numbers number of 0s.	7	K3	CO2&K2
12	Differentiate between checked and unchecked exceptions in Java with examples. Demonstrate the significance of the keywords 'try', 'catch', 'finally', 'throw' and 'throws' in exception handling of Java with appropriate examples.	7	K2	CO3&K2

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**ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.***(An ISO 9001:2015 Certified College)***DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING****B. TECH DEGREE SECOND INTERNAL TEST- FEBRUARY 2022****THIRD SEMESTER – Batch A&B****CST 205 : OBJECT ORIENTED PROGRAMMING USING JAVA**

Time: 2 Hours

Max. Marks: 50

PART A

Answer all questions (3x5=15 marks)

Qn. No.	Questions	Marks	K level	CO & KL
1	Draw and Explain the lifecycle of a thread showing the different states and methods invoked.	3	K2	CO4&K3
2	Show using java codes how three threads can be created using Runnable Interface.	3	K3	CO4&K3
3	List any five event sources and their corresponding event types and listeners used.	3	K2	CO2&K3
4	What is use of Swing Layout Manager class in java? Discuss any two Layout Managers.	3	K2	CO5&K3
5	Write a swing program to read an input from the user in a text box and display a message in Label if the number is an even number or odd number	3	K3	CO5&K3

PART B

Answer any five questions (7x5=35 marks)

Qn. No.	Questions	Marks	K level	CO & KL
6	Write a Java program to read integer numbers from a file and write prime numbers into the file prime.txt and composite numbers to composite.txt	7	K3	CO2&K3
7	How mouse events are handled in Java? Give suitable Java source code to illustrate the mouse actions.	7	K3	CO4&K3
8	What is Thread Synchronization? With an example illustrate the working of any one technique used for Thread synchronization in Java.	7	K2	CO4&K3
9	Discuss the event handling mechanism in Java using the Delegation Event Model?	7	K2	CO4&K3

enter
exit
press
click
release
move
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10	Discuss the various steps involved in the JDBC connection. Write sample code to establish database connectivity in Java.	7	K2	CO5&K3
11	Develop a GUI program that allows the user to input velocity in Kilometres/Hour, convert and display in Meters/Second. Use SWING controls and event handling techniques.	7	K3	CO5&K3
12	Write a Java program to store the marks of students corresponding to three subjects in database. The program should provide provision for the following. a) Create a table in database corresponding to a particular class. b) Find the topper of the class. c) Find the average class marks for a given subject	7	K3	CO5&K3



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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: CST205**Course Name: OBJECT ORIENTED PROGRAMMING USING JAVA**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions. Each question carries 3 marks*

Marks

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| ✓ | What is Just-In-Time compiler? | (3) |
| ✓ | Why Java is said to be a secure programming language? | (3) |
| ✓ | Does Java support multiple inheritance? Justify your answer. | (3) |
| ✓ | Why is the 'main' method in Java qualified as public, static, and void? | (3) |
| ✓ | Explain any three Byte Stream classes in Java. | (3) |
| ✓ | What are Checked Exceptions? Give an example. | (3) |
| ✓ | List any six Event Listener interfaces. | (3) |
| ✓ | Explain any three special string operations in Java | (3) |
| ✓ | Summarize any three features of Swing API. | (3) |
| ✓ | Differentiate between Components and Containers in Swing. | (3) |

PART B*Answer any one full question from each module. Each question carries 14 marks***Module 1**

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| 11 | a) Explain Lexical issues in Java with examples. | (6) |
| | b) Differentiate between function oriented and object oriented software design approaches using a suitable example. | (8) |
| 12 | a) Construct a UML Class diagram for Online Movie Ticket Booking System. The various entities involved in the system are Admin, Registered User, Visitor / Guest User, Movie, Book Ticket, Make Payment. | (8) |
| | b) Construct a UML Activity diagram for Food Ordering System, which shows the flows between the activity of Order, Delivery, Food Item, Category, Payment. | (6) |

Module 2

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| 13 | a) Explain different data types in Java. Give examples. | (8) |
| | b) Write a Java program to reverse bits of a given integer. | (6) |
| 14 | a) Demonstrate how objects are passed as function parameters with a suitable example. | (8) |
| | b) Write a Java program to find the frequency (count the occurrence) of each element in an integer array. | (6) |

Module 3

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| 15 | a) Develop a Java package named ' <u>evenpackage</u> ', with a class <u>Even</u> containing a static method that check whether a number is even or not, and returns that | (8) |
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information. Import this package in another class and use to check a number is even or not.

- b) Differentiate between *try-catch*, *throw* and *throws* keywords. (6)
- 16 a) Write a Java program that reads a binary file and write to another file. (8)
- b) Write Java code that reads a character file and prints the contents of file on the display, with a line number before each line. (6)

Module 4

- 17 a) Write a Java program to find the duplicate characters in a string. (8)
- b) What are the uses of *synchronized* keyword in Java? Explain with examples. (6)
- 18 a) Write a Java program that creates three threads. First thread generates a random positive number (>1) every 1 second. If the number is even, the second thread prints all even numbers between 1 and the generated number. If the number is odd, the third thread will print all odd numbers between 1 and the generated number. (10)
- b) Differentiate between Collection Interface and Collections Class. (4)

Module 5

- 19 a) How do you establish connection between a Java program and database? Explain the steps with sample code. (10)
- b) Write java code to demonstrate the execution of select and delete queries using JDBC (4)
- 20 a) How events are handled in java Swing? (4)
- b) Write a Java program using Swing to create a frame having three text fields, three labels and a button. The interface has to accept a number in the first text field. While clicking the button, the second and third textfields have to display the previous number and next number respectively, of the accepted input number. (10)

